

CAPSTONE PROJECT · BUSINESS REQUIREMENTS DOCUMENT

Mira: AI-Powered Project Intelligence Assistant

Business Requirements Document (BRD)

This BRD defines the business rationale for Mira — an AI assistant that automates project planning, risk assessment, and status reporting for Nexora Pvt. Ltd. It captures the users, the problem, business goals, success criteria, scope, and the grounding-and-trust requirements that make the system safe to rely on.

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Client / Context: Nexora Pvt. Ltd. (delivery partner) building for ABCDE Ltd. (end client)

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Contents

Section	Page
1. Executive Summary	3
2. Business Context & Problem Statement	3
3. Users & Stakeholders	4
4. Business Goals & Objectives	4
5. Success Criteria & Priority Metrics	5
6. Scope (In / Out)	5
7. Tool Choice & Rationale	6
8. Key Risks & Assumptions	6

1. Executive Summary

Nexora Pvt. Ltd., a 120-person London-based IT services company, has won the contract to deliver an **AI Adoption Project for ABCDE Ltd.**, a mid-sized logistics and supply-chain firm. Managing this engagement — and several others in parallel — the PM/TPM spends **3–4 hours creating each project plan by hand**, another **1–2 hours per week compiling status reports**, and produces **risk assessments inconsistently**. Milestones slip because problems surface only in retrospectives.

Mira is a multi-agent AI assistant that automates three core project-intelligence tasks — **project plan generation, risk assessment, and weekly status reporting** — each grounded strictly in the real ABCDE Ltd. project data (description, timeline, risk matrix, task board). Its defining requirement is **trust**: Mira must never fabricate milestones, task statuses, or risks, and must refuse to answer when input is too vague, rather than inventing a confident but false response.

2. Business Context & Problem Statement

2.1 Context

ABCDE Ltd. is adopting AI across demand forecasting, route optimization, inventory management, and customer support, targeting a 15% reduction in delivery delays and 10% lower operational costs over a 6-month, 8-phase programme. Nexora manages the delivery. The project generates continuous planning, risk, and status work that currently consumes the PM/TPM's time and varies in quality.

2.2 Problem Statement

The PM/TPM's project-intelligence workflow is manual, slow, and inconsistent:

Pain point	Current cost	Business impact
Creating project plans from high-level descriptions	3–4 hrs per project	Slow project starts; PM time lost to formatting
Compiling weekly status reports from the task board	1–2 hrs per week	Recurring drain; delayed visibility for leadership
Producing risk assessments	Done inconsistently	Some projects under-assessed; uneven governance
Detecting slipping milestones	Reactive only	Issues found late, in retrospectives

The core business risk Mira must solve for: a project assistant that *invents* information is worse than no assistant — a fabricated milestone or an incorrect task status feeds directly into decisions made with leadership and the client. Trust (grounding + refusal on vague input) is therefore the primary business requirement, not a nice-to-have.

3. Users & Stakeholders

Role	Who	Relationship to Mira
Primary user	PM / TPM at Nexora	Runs Mira daily to generate plans, risk assessments, and status reports
Business stakeholder	Nexora delivery lead / leadership	Consumes status reports and stakeholder updates; needs trustworthy, consistent output
End client	ABCDE Ltd. sponsors	Receive stakeholder updates derived from Mira's reports
Delivery team	Data scientists, ML/IT engineers (Arjun, Priya, Ravi, James, Sarah)	Task assignees whose work is reflected in the task board Mira reads

Note on roles for this capstone: as a solo capstone, the author performs every role in the SDLC — Business Stakeholder, Product Owner, Developer, Tester, Project Manager, and End User — and documents the project from each perspective. This BRD is authored from the Business Stakeholder / Product Owner viewpoint.

4. Business Goals & Objectives

#	Business goal	Objective (measurable)
G1	Cut time spent on project planning	Reduce plan creation from 3–4 hrs to minutes
G2	Make status reporting effortless and consistent	Generate a weekly status report on demand from the task board
G3	Standardize risk assessment quality	Every project gets a consistent, categorized risk matrix
G4	Surface at-risk milestones proactively	Flag blocked / at-risk items (e.g., T024) before retrospectives
G5	Guarantee trustworthy output	Zero fabricated facts; refuse vague input instead of inventing
G6	Deliver updates where stakeholders work	Push stakeholder updates via email (Gmail) and Slack

5. Success Criteria & Priority Metrics

Mira is judged not on how polished its output looks, but on whether that output is **grounded, correct, and trustworthy**. The priority metrics below (three categories, per the course rubric) define "working."

Category	Metric	Target
Accuracy	Groundedness — outputs trace to the source data (correct task counts, real milestones, actual risks)	Correct on all non-vague test cases (T1, T3, T5, T7, T8, T10, T11, T12)
Trustworthiness	Hallucination rate — % of outputs containing invented facts; correct refusal on vague inputs	Correct refusal on all vague cases (T2, T4, T6, T9); 0 fabricated facts
Usefulness	Time saved per plan/report; consistency across repeated runs	Plan/report in minutes; consistent output across 10 runs

The decisive success test: the 4 deliberately vague inputs (T2 "build a chatbot," T4 "new project starting soon," T6 "things are going fine," T9 "2-week project no details") must produce a **request for more information** — not a fabricated plan, risk list, or status. Passing these is the clearest proof Mira is trustworthy.

6. Scope

6.1 In Scope

- Three core capabilities: **Project Plan Generation, Risk Assessment, Weekly Status Report**
- A **Router** that classifies request type and dispatches to the right specialist agent
- Grounding in the four ABCDE Ltd. data files (description, timeline CSV, risk CSV, task board CSV)
- The "if vague → ask / UNKNOWN" guardrail in every agent
- **Delivery via Gmail and Slack** for stakeholder updates
- Baseline testing against all 12 provided inputs (T1–T12)

6.2 Out of Scope (for the mandatory pipeline)

- Live two-way sync with Trello/Jira (optional future enhancement)
- Fine-tuning (only if evaluation later proves prompting has plateaued — Phase 4/5)
- A production web UI beyond the workflow interface

7. Tool Choice & Rationale

Each tool is chosen with a stated "because," per the rubric.

Category	Choice	Because
Platform (mandatory)	n8n	Native Gmail and Slack nodes make the required delivery + integration trivial; strong at orchestrating a router + specialists; matches the author's prior project-status agents
Platform (secondary)	LangFlow	Built as a bonus second pipeline — native Langfuse tracing and the class-taught evaluation stack for deeper observability
LLM	gpt-4o-mini / 4.1-mini default	Cheap and fast for classify + grounded generation; low volume keeps cost ~\\$.20/user/month
APIs / Integration	File ingestion + Gmail + Slack	Reads the four project files; writes stakeholder updates to email and Slack — the delivery the business requires
Observability	Langfuse (+ n8n execution logs)	Trace per-agent input/output, tokens, latency, cost; run the baseline set and evaluate
Output	Markdown + email/ Slack messages	Portable, standardized format the team currently lacks

Why n8n over LangFlow as the mandatory pipeline: the business requires stakeholder updates to be *delivered* (Gmail + Slack), not just generated. n8n provides these as native, configure-in-clicks nodes, whereas LangFlow would require hand-coded integration. Since delivery is a stated business requirement, n8n is the pragmatic mandatory choice; LangFlow is delivered as a secondary pipeline to demonstrate the observability-first stack.

8. Key Risks & Assumptions

Risk / Assumption	Handling
Agent fabricates milestones, task statuses, or risks	Grounding rule in every prompt: use only provided data; cite from the CSVs
Agent answers vague input with invented detail	"If vague → ask for detail / UNKNOWN" guardrail; validated by T2/T4/T6/T9
Output varies run to run	Consistency checked across 10 runs in the evaluation phase
Assumes the four ABCDE data files are the single source of truth	All grounding traces back to these files; no external knowledge used
Delivery integrations (Gmail/Slack) require account auth	Configured in n8n; author performs setup as Developer role

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